# SET 7: THE LIGEIA MARE HYDROCARBON SUBSURFACE PIPELINE & LIQUEFACTION TERMINAL

## DOCUMENT 7B: BORROWER'S Q3 2146 QUARTERLY REPORTING PACKAGE & VALUATION MEMO

**To:** Jovian-Saturnine Colonial Development Bank (JSCDB), as Administrative Agent

**From:** Ligeia Transport & Liquefaction LP (Borrower)

**Date:** October 20, 2146

**Subject:** Q3 2146 Financial Compliance Report & Asset Valuation Analysis

### 1. Financial and Operational Narrative

Throughout the third quarter of 2146, the Ligeia Mare Hydrocarbon Subsurface Pipeline & Liquefaction facility operated in complete compliance with all covenants and technical metrics detailed in the Secured Credit Agreement (Document 7A). The piping network maintained strong structural performance, posting a quarterly average **Subsurface Pipeline Integrity Rating (SPIR)** of 95.4%, which exceeds the 90.0% baseline standard specified in Section 4.01(1).

Liquid hydrocarbon transport volumes averaged 9,485 metric tons of liquefied compounds per Titan sidereal day, successfully achieving the **Pipeline Throughput Target** of 9,200 metric tons outlined in Section 4.01(2). Consequently, the Facility Loans accrued interest at the baseline floating rate of **TIDOR plus 7.25%** per annum, with no triggers of the Pipeline Rupture Risk Premium. Additionally, our escrow balances in the **Undersea Sediment Escrow** were fully maintained at the required SC 450,000,000 under Section 4.01(3).

## 2. Financial Statements (SSIAS Compliant)

### Table 7.1: Consolidated Balance Sheet & Income Statement (Q3 2146)

All figures in Millions of Sol Credits (SC)

| BALANCE SHEET & INCOME STATEMENT CONSOLIDATION | | | |
|---|---|---|---|
| **Category / Account Item** | **Q3 2146** | **Q2 2146** | **Variance (%)** |
| **ASSETS** | | | |
| Cash & Cash Equivalents | 510.50 | 485.20 | +5.21% |
| Undersea Sediment Escrow[1] | 450.00 | 450.00 | +0.00% |
| Liquefied Methane Inventories | 710.10 | 675.00 | +5.20% |
| Pipeline & Liquefaction PP&E (Net)[2] | 5,600.00 | 5,720.00 | -2.10% |
| **TOTAL ASSETS** | **7,270.60** | **7,330.20** | **-0.81%** |
| **LIABILITIES & PARTNERS' CAPITAL** | | | |
| Secured Term Loans (JSCDB Facility)[3] | 4,800.00 | 4,950.00 | -3.03% |
| Benthic Remediation Provision | 820.00 | 815.00 | +0.61% |
| Partners' Equity Capital | 1,650.60 | 1,565.20 | +5.46% |
| **TOTAL LIABILITIES & CAPITAL** | **7,270.60** | **7,330.20** | **-0.81%** |
| **OPERATING INCOME STATEMENT** | | | |
| Gross Liquefied Methane Sales[4] | 812.50 | 785.10 | +3.49% |
| Marine Pipeline Ops & Liquefaction OpEx[5] | (345.30) | (341.00) | +1.26% |
| **Net Operating Income (NOI)** | **467.20** | **444.10** | **+5.20%** |

[1] Reconciles directly to the mandatory sediment stability reserve account specified in Section 4.01(3).

[2] Detailed asset class values are reconciled directly to the subtotal summation of Net Book Values reported in **Table 7.2**.

[3] Reconciles directly to the outstanding Facility Loan principal balance specified under Section 2.01.

[4] Annualized liquefied methane sales correlate directly with the Base Case model outlined in **Table 7.3**.

[5] Includes pipeline pressure monitoring, dynamic benthic anchoring, and cryogenic fractional plant maintenance.

### Table 7.2: Detailed Pipeline Property, Plant & Equipment (PP&E)

All figures in Millions of Sol Credits (SC)

| Asset Category | Sub-Asset Class | Gross Cost | Accumulated Depr. | Net Book Value |
|---|---|---|---|---|
| **1. Submarine Pipeline** | Titanium-Niobium Subsurface Pipeline | 2,600.00 | (550.00) | 2,050.00 |
| | Undersea Sediment Anchoring Anchor Arrays | 1,200.00 | (300.00) | 900.00 |
| | **Subtotal Subsurface Pipeline** | **3,800.00** | **(850.00)** | **2,950.00** |
| **2. Liquefaction Plant** | Cryogenic Fractional Separator Plant | 1,350.00 | (250.00) | 1,100.00 |
| | Methane Compression loops | 850.00 | (200.00) | 650.00 |
| | **Subtotal Liquefaction Plant** | **2,200.00** | **(450.00)** | **1,750.00** |
| **3. Marine Docking** | Submarine Docking Platform & Cables | 1,100.00 | (200.00) | 900.00 |
| | **Subtotal Docking Assets** | **1,100.00** | **(200.00)** | **900.00** |
| **TOTAL PIPELINE PP&E**[*] | | **7,100.00** | **(1,500.00)** | **5,600.00** |

[*] Reconciles directly to the **Pipeline & Liquefaction PP&E (Net)** line item on the Balance Sheet in **Table 7.1**.

## 3. Manager's Valuation Memo

**Valuation Methodology:** The Ligeia Mare Hydrocarbon Pipeline is valued utilizing a Discounted Cash Flow (DCF) model reflecting specific undersea sediment stability and cryogenic hydrostatic pressure variables. Key structural assumptions include a Weighted Average Cost of Capital (WACC) of 12.00% and processing centrifugal column mechanical wear cycles over a 20-year projected operational lifespan.

**Table 7.3: Net Asset Value (NAV) Modeling under Alternate Export Scenarios**

All figures in Millions of Sol Credits (SC) unless specified

| Valuation Parameter | Base Case (Methane @ SC 350/t) | Bear Case (Methane @ SC 260/t) | Bull Case (Methane @ SC 440/t) |
|---|---|---|---|
| **Annual Pipeline Throughput (k Tons)** | 9,285.71 | 8,500.00 | 10,000.00 |
| Gross Revenue (SC/yr)[#] | 3,250.00 | 2,210.00 | 4,400.00 |
| Marine Pipeline Utilities & Ops Costs | (1,381.20) | (1,420.00) | (1,340.00) |
| Pipeline Undersea Anchor Re-tensioning | (400.00) | (400.00) | (400.00) |
| **Net Operating Income (NOI)** | **1,468.80** | **390.00** | **2,660.00** |
| **Capitalized Value (8.5% Capitalization Rate)** | **17,280.00** | **4,588.24** | **31,294.12** |
| Debt Outstanding (JSCDB Facility)[^] | (4,800.00) | (4,800.00) | (4,800.00) |
| **Implied Net Asset Value (NAV)** | **12,480.00** | **(211.76)** | **26,494.12** |

[#] Pro-rata Q3 Gross Revenue matches the **Gross Liquefied Methane Sales** reported in **Table 7.1** (SC 812.50m represents exactly 25.0% of the annualized Base Case Gross Revenue of SC 3,250.00m).
[^] Corresponds to the **Secured Term Loans (JSCDB Facility)** balance reported in the liabilities section of **Table 7.1**.